

MMI 804 · MATHEMATICAL SYSTEM MODELLING AND SIMULATION

MSc in Mathematical Innovation · Open University of Kenya

Topic 9: Open Modelling

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LEARNING OUTCOMES

1. Document a modelling workflow in sufficient detail for independent peer verification and reproduction.
2. Use Git and GitHub to version control, share, and collaborate on simulation scripts.
3. Apply FAIR data principles (Findable, Accessible, Interoperable, Reusable) to modelling projects.
4. Critically evaluate the reproducibility of published simulation studies.

1 Topic Overview

Mathematical models are only as valuable as their credibility. A model whose code is unavailable, whose assumptions are undocumented, and whose results cannot be reproduced is a scientific liability. **Open Modelling** is the commitment to transparency, reproducibility, and community verification of mathematical models — a set of practices as important as the modelling mathematics itself.

THE REPRODUCIBILITY CRISIS

A 2016 *Nature* survey found that more than 70% of researchers had failed to reproduce another researcher's results. In computational modelling, the primary causes are: (1) code not shared; (2) code shared but undocumented; (3) random seeds not fixed; (4) software dependencies not specified; (5) parameter values not fully reported. Open modelling practices directly address all five.

2 The Open Modelling Manifesto

Open modelling commits to four principles:

1. **Open code:** Simulation scripts are published with the study.
2. **Open data:** Input data and calibration data are available.
3. **Open documentation:** Assumptions, parameter choices, and model structure are fully described.

4. **Version control:** Model evolution is tracked; any published version can be exactly reproduced.

3 Documenting a Modelling Workflow

3.1 The ODD Protocol

The **ODD Protocol** (Overview, Design Concepts, Details; Grimm et al., 2006) is the standard documentation framework for agent-based models:

ODD PROTOCOL — Structure

Overview:

1. Purpose — what question does the model address?
2. Entities, state variables, and scales
3. Process overview and scheduling

Design Concepts:

4. Basic principles (emergence, adaptation, objectives)
5. Learning; prediction; sensing; interaction; stochasticity; collectives; observation

Details:

7. Initialisation
8. Input data
9. Submodels

3.2 README-Driven Development for Models

For ODE/continuous models, use a structured README:

MODEL README TEMPLATE

```
# Model Name: [e.g., SIR_Kenya_Measles_2019]
## Purpose
One-sentence statement of what question this model answers.

## System Representation
- State variables: S, I, R (definitions and units)
- Parameters: beta, gamma (definitions, units, values, sources)
- Governing equations: [full system of ODEs]

## Assumptions
1. Homogeneous mixing (justification)
2. Closed population (justification)
...
```

```

## Software Requirements
- Python 3.11, scipy 1.11, numpy 1.26, matplotlib 3.8
- Install: pip install -r requirements.txt

## Reproducing Results
1. Run: python sir_model.py
2. Expected output: Figure 1 (epidemic curve), Table 1 (R0 estimates)
3. Runtime: ~30 seconds on standard laptop

## Data Sources
- Case data: Ministry of Health Kenya, 2019 (DOI: ...)
- Population: Kenya National Bureau of Statistics 2019 Census

```

4 Version Control with Git and GitHub

4.1 Why Version Control for Models?

Version control solves the “file hell” problem: `model_v1.py`, `model_v1_FINAL.py`, `model_v1_FINAL_rev1`, `model_v2_actually_final.py`. Git provides:

- Complete history of every change to every file
- Ability to roll back to any previous state
- Branching for experimental modifications without breaking the main model
- Collaboration without file conflicts
- A permanent, citable URL for any published model version

ACTIVITY: Git Workflow for a Modelling Project

Step 1: Initialise repository

```

mkdir sir_model && cd sir_model
git init
git add sir_model.py README.md requirements.txt
git commit -m "Initial commit: baseline SIR model"

```

Step 2: Push to GitHub

```

# Create repo at github.com, then:
git remote add origin https://github.com/username/sir_model
git push -u origin main

```

Step 3: Tag a published version

```

git tag -a v1.0 -m "Version used in MMI804 assignment 1"
git push origin v1.0

```

Step 4: Create a citable DOI via Zenodo

Connect GitHub repo to Zenodo (<https://zenodo.org>). Each GitHub release automatically generates a DOI for citation.

4.2 Good Practices for Reproducible Code

REPRODUCIBILITY CHECKLIST

1. **Fix random seeds:** `np.random.seed(42)` at the top of every stochastic script.
2. **Use requirements.txt:** `pip freeze > requirements.txt` — captures exact package versions.
3. **Use relative paths:** Never hardcode `C:/Users/name/project/data.csv`.
4. **Separate data from code:** Never modify raw data files; keep processed data in a separate folder.
5. **Comment functions:** Every function should have a docstring specifying inputs, outputs, and units.
6. **Use Jupyter notebooks sparingly:** For exploratory work only; convert to Python scripts for final runs.
7. **Log outputs:** Save figures with `plt.savefig('fig1.png', dpi=300)` — don't rely on screen capture.

5 FAIR Data Principles

FAIR PRINCIPLES (Wilkinson et al., 2016)

F — Findable: Data and metadata are assigned persistent identifiers (DOIs); indexed in searchable repositories.

A — Accessible: Data is retrievable via a standard protocol (e.g., HTTP); metadata remains accessible even if data is restricted.

I — Interoperable: Data uses standard formats (CSV, JSON, HDF5) and controlled vocabularies; linked to related data.

R — Reusable: Data is richly documented with provenance, licence (CC-BY, MIT), and standards-compliant metadata.

Repositories for modelling data: Zenodo, Figshare, OSF (Open Science Framework), Dryad (ecology), Harvard Dataverse.

6 Evaluating Reproducibility of Published Models

PEER EVALUATION ACTIVITY

Find a published agent-based or ODE-based modelling paper with accompanying code (e.g., from CoMSES Network: <https://www.comses.net> or GitHub). Evaluate it against:

1. **Availability:** Is the code publicly accessible? Is data available?

2. **Documentation:** Is there a README? Is the ODD protocol followed?
 3. **Reproducibility:** Can you run the code without modification and obtain the published figures?
 4. **Clarity:** Are parameters, units, and assumptions clearly stated?
 5. **Version control:** Is a specific version tagged or a DOI provided?
- Rate the paper on a 5-point open-science scale and write a 200-word justification.

7 Open Science and Ethical Responsibilities

Open modelling practices are not merely technical — they have ethical dimensions:

- **Public health:** Epidemic models that inform policy decisions must be scrutinisable by independent experts. Opaque models that cannot be reproduced should not inform policy.
- **Equity:** Open code and open data lower the barrier for researchers in low-resource settings to build on and critique existing work.
- **Accountability:** Models used in legal, financial, or governmental contexts should be auditable.
- **Scientific integrity:** Reproducibility is not just best practice — it is the foundation of the scientific method.

8 Key Concepts and Terminology

Reproducibility	Ability to re-obtain the same results from the same code and data.
ODD Protocol	Standard documentation framework for agent-based models.
Version Control	Systematic tracking of changes to files over time (Git).
FAIR Principles	Findable, Accessible, Interoperable, Reusable — data management standards.
DOI	Digital Object Identifier: persistent unique identifier for a digital resource.
Random Seed	Initial value for a pseudorandom number generator; fixing it ensures reproducibility.
Zenodo	Open-access repository for research outputs; assigns DOIs to GitHub releases.

9 Revision Questions

1. Take your SIR model from Topic 5 and apply the full open modelling workflow: write a README, create a requirements.txt, fix the random seed, push to GitHub, and generate a DOI via Zenodo.
2. Apply the ODD protocol to the ABM from Topic 6 (epidemic ABM in Mesa). Write a full ODD document of approximately 500 words.
3. Download a published ABM from CoMSES Network. Attempt to reproduce the main figure. Document any obstacles you encounter and how you resolved them.
4. What is the difference between *reproducibility* and *replicability* in scientific modelling? Why does this distinction matter for policy-relevant models?
5. A modelling team publishes COVID-19 predictions that later prove inaccurate. The code is not shared. What are the scientific, ethical, and policy consequences of this opacity?
6. Write unit tests in Python for the logistic growth ODE solver from Topic 1. What cases should you test? (Hint: test with $N_0 = K$, $r = 0$, and $N_0 \ll K$.)
7. Explain how Git branching can prevent a common modelling disaster: accidentally breaking a working model while exploring a modification. Describe a scenario and the Git commands you would use.

TOPIC 9 SUMMARY

- Open modelling — open code, open data, open documentation, version control — is a scientific and ethical responsibility, not an optional extra.
- The ODD protocol provides a standard framework for documenting agent-based models; similar structured README conventions serve ODE models.
- Git and GitHub enable reproducibility through version tracking, tagging, and permanent DOIs via Zenodo.
- FAIR principles ensure modelling data and code are findable, accessible, interoperable, and reusable.
- Reproducibility is the foundation of scientific credibility; models that inform policy must be independently auditable.
- *A model that cannot be reproduced is a hypothesis, not a result.*

10 Further Reading

- **Grimm, V. et al.** (2006). A standard protocol for describing individual- and agent-based models. *Ecological Modelling*, 198, 115–126. — ODD protocol.
- **Wilkinson, M.D. et al.** (2016). The FAIR Guiding Principles for scientific data management. *Scientific Data*, 3, 160018.
- **Wilson, G. et al.** (2014). Best Practices for Scientific Computing. *PLOS Biology*, 12(1).
- **CoMSES Net:** <https://www.comses.net> — Open access repository for computational models.